



GROUP MEMBERS NAME:

NASEER AHMAD (42273)

NAEEM ULLAH (42658)

M.ADIL (42657)

SUBMITTED TO:

MS. TAYYIBA SHEHZAD

SUBJECT:

COMPUTER ARCHITECTURE

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PROJECT REPORT

Voting Machine System Using Assembly Language (8086)

1. Introduction

This project is a simple **Voting Machine System** developed using **8086 Assembly Language**. The system allows users to cast votes for three different candidates and displays the final result with the winner.

The main purpose of this project is to understand low-level programming concepts such as memory handling, interrupts, loops, and conditional logic.

2. Objective

The objectives of this project are:

- To design a simple voting system in Assembly Language
- To understand input/output operations using interrupts
- To implement decision-making using comparison instructions
- To handle memory variables for vote counting
- To calculate and display the winner based on votes

3. Tools & Technology Used

- Assembly Language (8086)
- DOS Interrupts (INT 21H)
- Emulator (EMU8086 / DOSBox recommended)

4. Project Description

This system displays a menu with four options:

1. Candidate A
2. Candidate B
3. Candidate C
4. Show Result and Exit

The user enters a vote by pressing a number key. Each valid vote increases the respective candidate's counter.

If the user enters an invalid input, the system displays an error message and returns to the menu.

5. Working Methodology

The working process is as follows:

1. The program starts and displays the menu
2. User inputs a choice (1–4)
3. System checks input using **CMP instruction**
4. If valid:
 - Vote is stored using **INC instruction**
5. If invalid:
 - Error message is shown
6. If user presses **4**, result section is executed
7. Votes are compared and winner is displayed

6. Code Explanation (Key Parts)

Data Section

This section stores messages and variables:

- Menu strings
- Vote counters (VOTE_A, VOTE_B, VOTE_C)
- Result messages
- Winner messages

Input Handling

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```
MOV AH, 01H  
INT 21H
```

This takes a single character input from the user.

Vote Counting

```
INC BYTE PTR [VOTE_A]
```

This increases the vote count for Candidate A.

✦ Output Display

```
MOV AH, 09H  
INT 21H
```

Used to display strings on screen.

Comparison Logic

```
CMP AL, [VOTE_B]  
JB CHECK_B
```

Used to compare votes and decide the winner.

7. Output of the Program

Sample Run

1. Candidate A
2. Candidate B
3. Candidate C
4. Show Result and Exit

```
Enter Vote: 1  
Enter Vote: 1  
Enter Vote: 2  
Enter Vote: 3
```

Enter Vote: 1

Enter Vote: 4

Final Result

===== FINAL RESULT =====

Candidate A Votes: 3

Candidate B Votes: 1

Candidate C Votes: 1

Winner: Candidate A

8. Features of the Project

- Simple menu-driven system
- Real-time vote counting
- Input validation
- Winner calculation logic
- User-friendly DOS interface

9. Conclusion

This project successfully demonstrates a basic **Voting Machine System using Assembly Language**. It helps in understanding how low-level systems handle:

- Memory operations
- Input/output using interrupts
- Conditional logic
- Loop-based program flow

Overall, this project is very useful for learning system-level programming and hardware-level logic design.

10. Future Improvements

This system can be improved by:

- Adding password protection for admin
- Supporting more candidates
- Storing votes in external memory/file
- Creating graphical interface